

3.36 Let $H = 4\rho^2\mathbf{a}_\rho - 2z\mathbf{a}_z$. Verify the divergence theorem for the cylindrical region defined by $\rho = 10$, $0 < \phi < 2\pi$, $0 < z < 3$.

$$\Psi = \oint_V \mathbf{H} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{H} dv$$

To find $\oint_V \mathbf{H} \cdot d\mathbf{S}$, let

$$\Psi = \Psi_t + \Psi_b + \Psi_s$$

where Ψ_t , Ψ_b , and Ψ_s are the fluxes from the top, bottom, and side of the cylinder.

$$\text{For } \Psi_t, \quad d\mathbf{S} = \rho d\rho d\phi \mathbf{a}_z$$

$$\Psi_t = -2z \int_0^{10} \int_0^{2\pi} \rho d\rho d\phi \Big|_{z=3} = -6 \int_0^{10} \rho d\rho \int_0^{2\pi} d\phi = -12\pi \frac{\rho^2}{2} \Big|_0^{10} = -600\pi$$

$$\text{For } \Psi_b, \quad d\mathbf{S} = \rho d\rho d\phi (-\mathbf{a}_z)$$

$$\Psi_b = 2z \int_0^{10} \int_0^{2\pi} \rho d\rho d\phi \Big|_{z=0} = 0$$

$$\text{For } \Psi_s, \quad d\mathbf{S} = \rho d\phi dz \mathbf{a}_\rho$$

$$\Psi_s = 4\rho^3 \int_0^{10} \int_0^{2\pi} d\phi dz \Big|_{\rho=10} = 4000 \int_0^3 dz \int_0^{2\pi} d\phi = 4000(3)(2\pi) = 24000\pi$$

$$\Psi = -600\pi + 0 + 24000\pi = 23400\pi = \underline{73,513.27}$$

$$\text{To get } \int_V \nabla \cdot \mathbf{H} dv, \quad dv = \rho d\rho d\phi dz$$

$$\nabla \cdot \mathbf{H} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho H_\rho) = 2 - 2 = 0$$

$$\Psi = \int_0^3 \int_0^{2\pi} \int_0^{10} (2\rho - 2) \rho d\rho d\phi dz = 12 \int_0^3 \int_0^{2\pi} \rho^2 d\rho \int_0^{2\pi} d\phi dz - 2 \int_0^3 \int_0^{2\pi} \rho d\rho \int_0^{2\pi} d\phi dz$$

$$= 12 \left(\frac{1000}{3} \right) (2\pi) (3) - 2 \left(\frac{100}{2} \right) (2\pi) (3) = 24000\pi - 600\pi = \underline{73,513.27}$$

3.2 Calculate the areas of the following surfaces using the differential surface area dS :

- (a) $\rho = 2$, $0 < z < 5$, $\pi/3 < \phi < \pi/2$
 (b) $z = 1$, $1 < \rho < 3$, $0 < \phi < \pi/4$
 (c) $r = 10$, $\pi/4 < \theta < 2\pi/3$, $0 < \phi < 2\pi$
 (d) $0 < r < 4$, $60^\circ < \theta < 90^\circ$, $\phi = \text{constant}$

$$(a) \quad dS = \rho d\phi dz$$

$$S = \int dS = \rho \int \int d\phi dz = 2 \int_0^5 dz \int_{\pi/3}^{\pi/2} d\phi = 2(5) \left[\frac{\pi}{2} - \frac{\pi}{3} \right] = \frac{10\pi}{6} = \underline{5.236}$$

$$(b) \quad \text{In cylindrical, } dS = \rho d\rho d\phi$$

$$S = \int dS = \int_1^3 \int_0^{\pi/4} \rho d\rho d\phi = \frac{\pi}{4} \left[\frac{\rho^2}{2} \right]_1^3 = \underline{3.142}$$

$$(c) \quad \text{In spherical, } dS = r^2 \sin\theta d\theta d\phi$$

$$S = \int dS = 100 \int_0^{2\pi} \int_{\pi/4}^{2\pi/3} \sin\theta d\theta d\phi = 100(2\pi)(-\cos\theta) \Big|_{\pi/4}^{2\pi/3} = 200\pi(0.5 + 0.7071) = \underline{758.4}$$

$$(d) \quad dS = r dr d\theta$$

$$S = \int dS = \int_0^4 \int_{\pi/6}^{\pi/4} r dr d\theta = \frac{r^2}{2} \Big|_0^4 \Big|_{\pi/6}^{\pi/4} = \frac{8\pi}{6} = \underline{4.189}$$

3.13 Let $\mathbf{A} = y\mathbf{a}_x + z\mathbf{a}_y + x\mathbf{a}_z$. Find the flux of $\mathbf{A}$ through surface $y = 1$, $0 < x < 1$, $0 < z < 2$.

$$\Psi = \int \mathbf{A} \cdot d\mathbf{S} = \int \int \int z dx dz = \int_0^2 \int_0^1 z dz = (1) \frac{z^2}{2} \Big|_0^2 = \underline{2}$$

3.15 A vector field is specified as $\mathbf{A} = r\mathbf{a}_r - 3a_\theta + 5\phi\mathbf{a}_\phi$. Find the flux of the field out of the closed surface defined by $0 < r < 4$, $0 < \theta < \pi/2$, $0 < \phi < 3\pi/2$.

$$\Psi = \oint_V \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} dv$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (-3 \sin\theta) + \frac{1}{r \sin\theta} (5) = 3 - \frac{3 \cos\theta}{r \sin\theta} + \frac{5}{r \sin\theta}$$

$$dv = r^2 \sin\theta dr d\theta d\phi$$

$$\Psi = 3 \int \int \int r^2 \sin\theta d\theta d\phi dr - 3 \int \int \int r \cos\theta d\theta d\phi dr + \int \int \int 5 d\theta d\phi dr$$

$$= 3 \int_0^4 r^2 dr \int_0^{\pi/2} \sin\theta d\theta \int_0^{3\pi/2} d\phi - 3 \int_0^4 r dr \int_0^{\pi/2} \cos\theta d\theta \int_0^{3\pi/2} d\phi + 5 \int_0^4 dr \int_0^{\pi/2} d\theta \int_0^{3\pi/2} d\phi$$

$$= 3 \left(\frac{r^3}{3} \Big|_0^4 \right) (-\cos\theta) \Big|_0^{\pi/2} (3\pi/2) - 3 \left(\frac{r^2}{2} \Big|_0^4 \right) (\sin\theta) \Big|_0^{\pi/2} (3\pi/2) + 5(4)(\pi/2)(3\pi/2)$$

$$= 96\pi - 36\pi + 15\pi^2 = \underline{336.54}$$

4.5 Determine the total charge

$$(a) \quad \text{On line } 0 < x < 5 \text{ m if } \rho_L = 12x^2 \text{ mC/m}$$

$$(b) \quad \text{On the cylinder } \rho = 3, 0 < z < 4 \text{ m if } \rho_S = \rho z^2 \text{ nC/m}^2$$

$$(c) \quad \text{Within the sphere } r = 4 \text{ m if } \rho_V = \frac{10}{r \sin\theta} \text{ C/m}^3$$

$$(a) \quad Q = \int \rho_L dl = \int_0^5 12x^2 dx = 4x^3 \Big|_0^5 = \underline{0.5 C}$$

$$(b) \quad Q = \int \rho_S dS = \int_{z=0}^4 \int_{\phi=0}^{2\pi} \rho z^2 d\phi dz = 9(2\pi) \frac{z^3}{3} \Big|_0^4 = \underline{1.206 \mu C}$$

$$(c) \quad Q = \int \rho_V dV = \int \int \int \frac{10}{r \sin\theta} r^2 \sin\theta d\theta d\phi dr$$

$$= 10 \int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \int_0^4 r dr = 10(2\pi)(\pi) \frac{4^2}{2}$$

$$= \underline{1579.1 C}$$

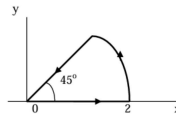
3.42 Let $\mathbf{H} = \rho \sin\phi \mathbf{a}_\rho + \rho \cos\phi \mathbf{a}_\phi - \rho \mathbf{a}_z$; find $\nabla \times \mathbf{H}$ and $\nabla \cdot \nabla \times \mathbf{H}$.

$$\nabla \times \mathbf{H} = 0\mathbf{a}_\rho + 1\mathbf{a}_\phi + \frac{1}{\rho} (2\rho \cos\phi - \rho \cos\phi) \mathbf{a}_z = \underline{\mathbf{a}_\phi + \cos\phi \mathbf{a}_z}$$

$$\nabla \cdot \nabla \times \mathbf{H} = \left(-\frac{1}{\rho} \sin\phi - 0 \right) \mathbf{a}_\rho + 0\mathbf{a}_\phi + \frac{1}{\rho} (1 - 0) \mathbf{a}_z = -\frac{1}{\rho} \sin\phi \mathbf{a}_\rho + \frac{1}{\rho} \mathbf{a}_z$$

3.46 If $\mathbf{F} = 2\rho z\mathbf{a}_\rho + 3z \sin\phi \mathbf{a}_\phi - 4\rho \cos\phi \mathbf{a}_z$, verify Stokes's theorem for the open surface defined by $z = 1$, $0 < \rho < 2$, $0 < \phi < 45^\circ$.

Prob. 3.46



$$\oint \mathbf{F} \cdot d\mathbf{l} = \int_0^2 2\rho z d\rho \Big|_{z=1} + \int_0^{45^\circ} 3z \sin\phi \rho d\phi \Big|_{\rho=2, z=1} + \int_0^2 2\rho z d\rho \Big|_{z=1}$$

$$= \rho^2 \Big|_0^2 + (-6 \cos\phi) \Big|_0^{45^\circ} + \rho^2 \Big|_0^2 = (4 - 0) + 6(-\cos\pi/4 + 1) + (0 - 4) = \underline{1.757}$$

$$\nabla \times \mathbf{F} = \frac{1}{\rho} [3z \sin\phi - 0] \mathbf{a}_z + \dots$$

$$\int (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_0^2 \int_0^{45^\circ} 3z \sin\phi \rho d\phi d\rho \Big|_{z=1} = 3(2)(-\cos\phi) \Big|_0^{45^\circ}$$

$$= 6(-\cos\pi/4 + 1) = \underline{1.757} \quad \text{4 of 4}$$

4.6 A cube is defined by $0 < x < a$, $0 < y < a$, and $0 < z < a$. If it is charged with

$$\rho_v = \frac{\rho_0 x}{a}, \text{ where } \rho_0 \text{ is a constant, calculate the total charge in the cube.}$$

4.7 A volume charge with density $\rho_v = 5\rho^2 z \text{ mC/m}^3$ exists in a region defined by $0 < \rho < 2$, $0 < z < 1$, $30^\circ < \phi < 90^\circ$. Calculate the total charge in the region.

Prob. 4.6

$$Q = \int \rho_v dv = \int_0^a \int_0^a \int_0^a \frac{\rho_0 x}{a} dx dy dz = (a)(a) \rho_0 \left(\frac{x^2}{2a} \Big|_0^a \right) = \underline{\frac{a^3 \rho_0}{2}}$$

Prob. 4.7

$$Q = \int \rho_v dv = \int_{\rho=0}^2 \int_{z=0}^1 \int_{\phi=0}^{\pi/2} 5\rho^2 z \rho d\phi d\rho dz \text{ mC}$$

$$= 5 \frac{\rho^4}{4} \Big|_0^2 \frac{z^2}{2} \Big|_0^1 \phi \Big|_0^{\pi/2} = \frac{5}{8} (16)(1)(\pi/2 - \pi/6) = \frac{10\pi}{3}$$

$$Q = \underline{10.472 \text{ mC}}$$